

Introduction to Ab Initio Nuclear Theory

Lecture 5: The No-Core Shell Model

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1 Introduction to the harmonic oscillator basis

1.1 Definitions

1.1.1 The coordinates

In this lecture we're concerned solely about the description of nuclei, i.e., *systems of many (more than two) particles self-bound by many- (two- and more) particle forces of finite range*. That is, the methods described here cannot be used (as are at least) for the description of systems in the continuum. In other words, the wavefunction of these systems is localized about a centre-of-mass whose motion is irrelevant to us. In this section we will establish the language used to describe them.

Throughout this Lecture we'll refer to two types of coordinates to describe an N -body system in three dimensions: the Cartesian coordinates and the Jacobi coordinates. A complete description of the system can be achieved in either coordinate basis but each has its own advantages and disadvantages. There is nothing "quantum" about this section: the same coordinates are used in celestial mechanics. We will be setting the language.

Cartesian Coordinates These we'll label as $\mathbf{r}_i$ for the i -th particle. In the language of the previous Lecture, we can bundle all coordinates in a single vector (which then we called "walker"; here we will not):

$$\mathbf{R} = (r_{11}, r_{12}, \dots, r_{1d}, \dots, r_{N1}, r_{N2}, \dots, r_{Nd})^T. \quad (1)$$

In this Lecture we'll also deal a lot with momenta (you might already suspect this given the HO's democratic treatment of momentum and space coordinates). These we'll label $\mathbf{p}_i$ for the i -th particle and their bundled vector as

$$\mathbf{P} = (p_{11}, p_{12}, \dots, p_{1d}, \dots, p_{N1}, p_{N2}, \dots, p_{Nd})^T, \quad (2)$$

Jacobi coordinates There is many ways to define the Jacobi coordinates. We choose:

$$\boldsymbol{\xi}_0 = \sqrt{\frac{1}{N}} \left(\sum_{i=1}^N \mathbf{r}_i \right), \quad (3)$$

$$\boldsymbol{\xi}_n = \sqrt{\frac{n}{n+1}} \left[\frac{1}{n} \sum_{i=1}^n \mathbf{r}_i - \mathbf{r}_{n+1} \right] \quad \text{with } n = 1, N-1 \quad (4)$$

The coordinate $\boldsymbol{\xi}_0$ is proportional to the COM of the system and each $\boldsymbol{\xi}_n$ is proportional to the distance between particle $n+1$ and the COM of particles $1, 2, \dots, n$. Jacobi coordinates are natural for decoupling the COM motion which depends solely on $\boldsymbol{\xi}_0$.

As before we can compose a vector out of these

$$\boldsymbol{\Xi} = (\xi_{11}, \xi_{12}, \dots, \xi_{1d}, \dots, \xi_{N1}, \xi_{N2}, \dots, \xi_{Nd})^T, \quad (5)$$

where again we mean $\boldsymbol{\xi} = (\xi_{11}, \xi_{12}, \dots, \xi_{1d})$. One can then relate the Jacobi to the Cartesian coordinates as

$$\boldsymbol{\Xi} = \Gamma \mathbf{R} \quad (6)$$

Since they are both equivalent descriptions of the many-body motion the transformation between $\mathbf{R}$ and $\boldsymbol{\Xi}$ ought to be a unitary one (i.e., $\Gamma \Gamma^{-1}$) and one can find the elements of the transformation as $\Gamma_{\xi_{ij} r_{kl}} = \nabla_{r_{kl}} \xi_{ij}$. Since unitary the transformation is also invertible, i.e., $\mathbf{R} = \Gamma^{-1} \boldsymbol{\Xi}$ providing a relation between Cartesian and Jacobi coordinates:

$$\mathbf{r}_n = \sqrt{\frac{1}{N}} \boldsymbol{\xi}_0 - \sqrt{\frac{n-1}{n}} \boldsymbol{\xi}_{n-1} + \sum_{m=n}^{N-1} \sqrt{\frac{1}{m(m+1)}} \boldsymbol{\xi}_m \quad (7)$$

Our choice for the definition of the Jacobi coordinate in Eqs. (3) and (4) makes it so that the canonical momenta, i.e., the Jacobi momenta are now related to the Cartesian ones in the same way as the spacial coordinates:

$$\boldsymbol{\pi}_0 = \sqrt{\frac{1}{N}} \left(\sum_{i=1}^N \mathbf{p}_i \right), \quad (8)$$

$$\boldsymbol{\pi}_n = \sqrt{\frac{n}{n+1}} \left[\frac{1}{n} \sum_{i=1}^n \mathbf{p}_i - \mathbf{p}_{n+1} \right] \quad \text{with } n = 1, N-1 \quad (9)$$

or in other words, the bundle

$$\boldsymbol{\Pi} = (\pi_{11}, \pi_{12}, \dots, \pi_{1d}, \dots, \pi_{N1}, \xi_{N2}, \dots, \pi_{Nd})^T, \quad (10)$$

is related to $\mathbf{P}$ as

$$\boldsymbol{\Pi} = \Gamma \mathbf{P} \quad (11)$$

with the same transformation matrix Γ meaning that the reverse transformation is identical too

$$\mathbf{p}_n = \sqrt{\frac{1}{N}} \boldsymbol{\pi}_0 - \sqrt{\frac{n-1}{n}} \boldsymbol{\pi}_{n-1} + \sum_{m=n}^{N-1} \sqrt{\frac{1}{m(m+1)}} \boldsymbol{\pi}_m \quad (12)$$

1.2 The single particle Harmonic Oscillator basis

We've already discussed how a many-body basis is directly related to a one-body one. So devising an efficient many-body basis for an expansion method depends heavily on the choice of the single-particle states. Let's review then the properties of the single-particle HO basis:

$$\phi_{nl}(\mathbf{x}; b) = R_{nl}(|\mathbf{x}|; b) [Y_{lm}(\hat{\mathbf{x}})] \quad (13)$$

$$= \left[\frac{2b^{l+3/2}n!}{\Gamma(l+n+3/2)^3} \right]^{\frac{1}{2}} |\mathbf{x}|^l e^{-b|\mathbf{x}|^2/2} L_n^{l+1/2}(b|\mathbf{x}|^2) [Y_{lm}(\hat{\mathbf{x}})] \quad (14)$$

where $\sqrt{b} = \hbar/m\omega$ and $n+1$ counts the number of nodes in the single-particle state. Each of these single-particle states corresponds to an eigenstate of the HO with eigenvalue

$$E_{nl} = \left(2n + l + \frac{3}{2} \right) \hbar\omega \quad (15)$$

1.3 Why (not) the HO basis: The good and the bad

In this lecture we're concerned solely about the description of nuclei, i.e., *systems of many (more than two) particles self-bound by many- (two- and more) particle forces of finite range*. This endeavour goes by the name of "nuclear structure" and here we list what the good, the bad, and the ugly of it: what one gains, what one loses (short comings introduced that take compute to correct which would have been absent in other bases which don't have the good), and has to live (complications that are not easily solved by other bases) with when using HO orbitals.

1.3.1 The good

There are many reasons that favour the HO basis for nuclear structure computations. Some of them have to do with the fact that the HO is quadratic in positions and the kinetic energy is quadratic in momentum (centre of mass factorization, analytical properties) and others with its infinite depth (discrete basis set).

They are analytical Their analytical nature allows for easy and cheap computation of integrals and transformations

They posses exact factorization of the centre-of-mass motion

They are a complete *discrete* basis Basis generated from finite-depth potentials contain a continuous piece that is hard to handle.

They are localized So their densities and currents are also localized

1.3.2 The bad

With the good things listed in the previous paragraph come also with some shortcomings that could be easily solved with other bases in the expense of all the *good*.

They have the wrong asymptotics The HO wavefunctions drop as gaussians at large distances $e^{\alpha x^2}$ while systems self-bound with finite-range forces have exponentially decaying wavefunctions. One can in principle approximate an exponential tail as a sum of many Gaussian ones. In quantum chemistry this is encountered when describing Slater-type orbitals (which exponentially decay) with Gaussian orbitals.

The fact that many Gaussian tails are needed to describe an exponential one makes the calculations expensive when dealing with weakly-bound systems (these have large radii) or with quantities sensitive to the systems size / radius.

2 The Hamiltonian

We are again concerned with solving the *time-independent* Schrödinger's equation $H |\Psi_n\rangle = E |\Psi_n\rangle$. In the case of the NCSM, where one diagonalizes the Hamiltonian directly, the access to the excited states is straightforward. We recast the many nucleon Hamiltonian in a favorable form

$$H_{\text{rel}} = T_{\text{rel}} + V_{NN} + V_{NNN} \quad (16)$$

$$= \frac{1}{A} \sum_{i < j}^A \frac{1}{2m} (\mathbf{p}_i - \mathbf{p}_j)^2 + \sum_{i < j}^A V_{NN}(\mathbf{r}_i, \mathbf{r}_j) + \sum_{i < j}^A V_{NNN}(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k) \quad (17)$$

As before we include up to three-nucleon forces with the inclusion of four-, five-, etc., nucleon forces being straightforward. Note that here we have by construction decoupled the centre-of-mass motion as we're interested in self-bound systems in vacuum and we'll be ignoring any COM excitations. Furthermore, while not explicitly shown, it is assumed that the two- and three-nucleon forces do not depend on the COM of the two-nucleon and the three-nucleon system, respectively. This is the case for all nuclear interactions.

One can formulate the problem of self-bound particles both with relative coordinates or Cartesian ones. The former description is more natural as it factors the COM movement, but it becomes impractical as one moves to heavier systems. This is because with Jacobi coordinates the basis used to describe the many-body Hilbert space of N particles is substantially different from that of $N + 1$ particles; we'll see this and its implications in practice below. When Cartesian coordinates one uses the so-called Lawson projection to ensure that the COM is not excited whereby one adds the Lawson projection term,

$$H_L = \beta(H_{\text{CM}} - 3/2\hbar\omega) \quad (18)$$

where the COM Hamiltonian H_{CM} is

$$H_{\text{CM}} = T_{\text{CM}} + U_{\text{CM}} = T_{\text{CM}} + \frac{1}{2}Am\omega \left(\frac{1}{A} \sum_i \mathbf{r}_i \right)^2 \quad (19)$$

where ω is the frequency of the Harmonic oscillator that we use to generate the single-particle orbitals. This is a good place to note that *there is no HO potential in the Hamiltonian*. The HO is used only to generate the single-particle orbitals. The Lawson term is added to the Hamiltonian and the parameter β is tuned to shift COM excitations to very high energies. We will give both formulations below.

3 The bases

3.0.1 The many-body basis with Jacobi coordinates (*Jacobi basis*)

The main distinction between the two bases is the how the antisymmetrization is handled. In fact, this distinction is what makes the Jacobi basis impractical for heavier systems. In this section will show how to do this for three particles and only hint towards the generalization to $A \geq 4$.

For $A = 3$, the relevant Jacobi coordinates, in temrs of the Cartesian coordinates $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$, are

$$\xi_0 = \sqrt{\frac{1}{3}}(\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3) , \quad \xi_1 = \sqrt{\frac{1}{2}}(\mathbf{r}_1 - \mathbf{r}_2) , \quad \xi_2 = \sqrt{\frac{2}{3}} \left[\frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2) - \mathbf{r}_3 \right] \quad (20)$$

As mentioned above, the explicit dependence on the centre of mass $\mathbf{R} = \xi_0/\sqrt{3}$ makes it possible to ignore it immediately (also knowing that it factorizes exactly in the many-body basis); this is the advantage of this basis. Hence, from now on we will only deal with coordinates ξ_1 and ξ_2 . The eigenstates of a rotationally invariant system in the COM frame are

$$\langle \xi_1 \xi_2 | (nlsjt; \mathcal{N}\mathcal{L}\mathcal{S}\tau\mathcal{J})JT \rangle , \quad (21)$$

where n, l and $\mathcal{N}, \mathcal{L}$ are the HO quantum number associated with the coordinates ξ_1 and ξ_2 respectively, and s, t and $\mathcal{S}$ and τ are the total spin and isospin of particles 1 and 2 and the spin and isospin of particle 3. In more detail, n and l are quantum numbers that describe the relative motion of particles 1 and 2, s and t are the total spin and isospin of particles 1 and 2, i.e., $s = s_1 + s_2$, $t = t_1 + t_2$, and $j = l + s$; that is, the quantum numbers $nlsjt$ all refer to the two-body system of particles 1 and 2. The quantum numbers $\mathcal{N}$ and $\mathcal{L}$ describe the relative motion of particle 3 from the COM of particles 1 and 2, $\mathcal{S} = 1/2$ and $\tau = 1/2$ are the spin and isospin of particle 3, and $\mathcal{J} = \mathcal{L} + \mathcal{S}$; that is, the quantum numbers $\mathcal{N}\mathcal{L}\mathcal{S}\mathcal{J}\tau$ all refer to particle 3 *measured from the COM of particles 1 and 2, not the total COM* (so not really a one-body system). So l is the orbital relative angular momentum of particles 1 and 2 and $\mathcal{L}$ is the orbital angular momentum of particle 3 with respect to the COM of particles 1 and 2, similarly j is the total angular momentum of particles 1 and 2 and $\mathcal{J}$ is the total angular momentum of particle 3. Finally, T is the system's total isospin while J is the system's total angular momentum. All the above quantum numbers, with the exception of n and $\mathcal{N}$, have z -projections which describe the subsystem's orientation and their coupled together to produce T and J . The z -projections of J and T , that is, J_z and T_z describe the geometric orientation of the *system as a whole* and are thus irrelevant, which is why we skipped them in Eq. (21) (but we'll show them below). To be maximally explicit we'll also provide the composition of the state in Eq. (21) in terms of individual angular momenta

“peeling off” the coupling layers:

$$\begin{aligned}
|(nlsjt; \mathcal{N}\mathcal{L}\mathcal{S}\tau\mathcal{J})JJ_zTT_z\rangle &= \sum_{\substack{t_z\tau_z \\ j_zJ_z}} C_{t_z\tau_zT_z}^{t\tau T} C_{j_zJ_zJ_z}^{j\mathcal{J}J} |n(ls)jj_ztt_z; \mathcal{N}(\mathcal{L}\mathcal{S})\mathcal{J}\mathcal{J}_z\tau\tau_z\rangle \\
&= \sum_{\substack{j_zJ_z \\ l_zs_z \\ \mathcal{L}_z\hat{S}_z}} C_{j_zJ_zJ_z}^{j\mathcal{J}J} C_{l_zs_zj_z}^{lsj} C_{\mathcal{L}_z\hat{S}_zJ_z}^{\mathcal{L}\frac{1}{2}\mathcal{J}} |nll_zss_ztt_z; \mathcal{N}\mathcal{L}\mathcal{L}_z\mathcal{S}\mathcal{S}_z\tau\tau_z\rangle .
\end{aligned} \tag{22}$$

and so

$$\langle \xi_1 \xi_2 | (nlsjt; \mathcal{N}\mathcal{L}\mathcal{S}\tau\mathcal{J})JJ_zTT_z \rangle = \sum_{\substack{j_zJ_z \\ l_zs_z \\ \mathcal{L}_z\hat{S}_z}} C_{j_zJ_zJ_z}^{j\mathcal{J}J} C_{l_zs_zj_z}^{lsj} C_{\mathcal{L}_z\hat{S}_zJ_z}^{\mathcal{L}\frac{1}{2}\mathcal{J}} \langle \xi_1 \xi_2 | nll_zss_ztt_z; \mathcal{N}\mathcal{L}\mathcal{L}_z\mathcal{S}\mathcal{S}_z\tau\tau_z \rangle \tag{23}$$

$$= \sum_{\substack{j_zJ_z \\ l_zs_z \\ \mathcal{L}_z\hat{S}_z}} C_{j_zJ_zJ_z}^{j\mathcal{J}J} C_{l_zs_zj_z}^{lsj} C_{\mathcal{L}_z\hat{S}_zJ_z}^{\mathcal{L}\frac{1}{2}\mathcal{J}} R_{nl}(\xi_1) R_{\mathcal{N}\mathcal{L}}(\xi_2) Y_{ll_z}(\hat{\xi}_1) Y_{\mathcal{L}\mathcal{L}_z}(\hat{\xi}_2) \chi_{s_z}(s) \chi_{\mathcal{S}_z}\left(\frac{1}{2}\right) \eta_{t_z}(t) \eta_{\tau_z}\left(\frac{1}{2}\right) \tag{24}$$

Implicit to these equations is that this basis is not fully antisymmetric, but only antisymmetric for particles 1 and 2. Exchanging the positions of particles 1 and 2, i.e., $\xi_1 \rightarrow -\xi_1$, will yield a the same state multiplied by a phase $(-1)^l$, coming from the parity of the spatial part $R_{nl}(\xi_1)$, and the phases $(-1)^s$ from the CG coefficient hidden in $\chi_{s_z}(s)$ and $(-1)^t$ from the CG coefficient hidden in $\eta_{s_z}(s)$; then the quantum numbers are chosen such that $(-1)^{l+s+t} = -1$. All together

$$\langle -\xi_1 \xi_2 | (nlsjt; \mathcal{N}\mathcal{L}\mathcal{S}\tau\mathcal{J})JJ_zTT_z \rangle = (-1)^{l+s+t} \langle \xi_1 \xi_2 | (nlsjt; \mathcal{N}\mathcal{L}\mathcal{S}\tau\mathcal{J})JJ_zTT_z \rangle = (-1) \langle \xi_1 \xi_2 | (nlsjt; \mathcal{N}\mathcal{L}\mathcal{S}\tau\mathcal{J})JJ_zTT_z \rangle \tag{25}$$

However, no such relation holds between particles 1 and 3 or 2 and 3, in other words, only particles 1 and 2 are indistinguishable. Hence, in practice, one performs calculations on the basis of Eq. (21) and then transforms any matrix elements calculated in a completely antisymmetrized basis.

The completely antisymmetric basis The completely antisymmetric basis is defined by diagonalizing the antisymmetrizer,

$$\mathcal{A}_3 = \frac{1}{3} (1 - P_{12} - P_{13} - P_{23} + P_{123} + P_{132}) \tag{26}$$

where P_{ij} describes the one-permutation of coordinate indices for particles i and j , i.e., $i \leftrightarrow j$ and P_{ijk} describes a two-permutation in the cyclic notation, i.e., $i \rightarrow j \rightarrow k \rightarrow i$. Note that this is the same antisymmetrizer that we saw in Lecture 4 which turned the product of orbitals into a Slater Determinant. For the three-nucleon system it can be written as

$$\mathcal{A}_3 = (1 - P_{ij})(1 + P_{123} + P_{132}) \tag{27}$$

for any choice of ij . If we choose $ij = 12$, then for two states $|\alpha\rangle$ and $|\alpha'\rangle$ that belong to the basis in Eq. (21), and so are antisymmetric in exchanges of particles 1 and 2, i.e., $(1 - P_{12})|\alpha\rangle = |\alpha\rangle$, it is

$$\langle \alpha' | \mathcal{A}_3 | \alpha \rangle = \langle \alpha' | (1 + P_{123} + P_{132}) | \alpha \rangle \tag{28}$$

The matrix elements of the two-permutation operators are equal because

$$P_{123} = P_{12}P_{23} , \quad P_{132} = P_{13}P_{23} = P_{12}P_{23}P_{12}P_{23} = P_{12}(P_{12}P_{23}P_{12}) = P_{23}P_{12} , \tag{29}$$

and so

$$\langle \alpha' | P_{123} | \alpha \rangle = \langle \alpha' | P_{12} P_{23} | \alpha \rangle = - \langle \alpha' | P_{23} | \alpha \rangle \quad (30)$$

$$\langle \alpha' | P_{132} | \alpha \rangle = \langle \alpha' | P_{23} P_{12} | \alpha \rangle = - \langle \alpha' | P_{23} | \alpha \rangle \quad (31)$$

Finally, the matrix elements of the antisymmetrizer on the Jacobi basis are

$$\langle \alpha' | \mathcal{A}_3 | \alpha \rangle = \frac{1}{3} (1 - 2 \langle \alpha' | P_{23} | \alpha \rangle) . \quad (32)$$

So calculating the matrix elements of the antisymmetrizer involves only calculating the matrix elements of the operator P_{23} . Intuitively, this makes sense since any transposition can be expressed in terms of P_{12} and P_{23} , and the former is trivial. The relevant matrix element is

$$\langle \alpha' | P_{23} | \alpha \rangle = \langle (n'l's'j't'; \mathcal{N}'\mathcal{L}'\mathcal{S}'\mathcal{T}'\mathcal{J}')JT | P_{23} | (nlsjt; \mathcal{N}\mathcal{L}\mathcal{S}\mathcal{T}\mathcal{J})JT \rangle \quad (33)$$

$$= \delta_{N'N} \hat{t}\hat{t}' \left\{ \begin{array}{ccc} \frac{1}{2} & \frac{1}{2} & t' \\ \frac{1}{2} & T & t \end{array} \right\} \sum_{LS} \hat{L}^2 \hat{S}^2 \hat{j}\hat{j}' \hat{\mathcal{J}}\hat{\mathcal{J}}' \hat{s}\hat{s}' \times \\ \times (-1)^L \left\{ \begin{array}{ccc} l & s & j \\ \mathcal{L} & \frac{1}{2} & \mathcal{J} \\ L & S & J \end{array} \right\} \left\{ \begin{array}{ccc} l' & s' & j' \\ \mathcal{L}' & \frac{1}{2} & \mathcal{J}' \\ L & S & J \end{array} \right\} \left\{ \begin{array}{ccc} \frac{1}{2} & \frac{1}{2} & s' \\ \frac{1}{2} & S & s \end{array} \right\} \langle n'l'\mathcal{N}'\mathcal{L}'L | \mathcal{N}\mathcal{L}nlL \rangle_3 \quad (34)$$

where $N = 2n + l + 2\mathcal{N} + \mathcal{L}$ and $\langle n'l'\mathcal{N}'\mathcal{L}'L | \mathcal{N}\mathcal{L}nlL \rangle_3$ is the overlap of HO wavefunction for two particles of mass ratio 3. The antisymmetrizer in its practical form in Eq. (32) has eigenstates spanning two eigenspaces, those that correspond to completely antisymmetric states, also called *physical states*, which come with eigenvalue 1, and so-called spurious states which are not antisymmetric and come with eigenvalue 0. They are about two times more in number than the physical ones and must be disregarded. Since the physical states correspond to eigenvalue 1, their number, namely r , can be easily calculated as the trace of the antisymmetrizer (remember the trace is invariant under unitary transformations): $r = \text{Tr}(\mathcal{A}_3)$.

The physical eigenstates, $|NiJT\rangle$, with $i = 1, 2, \dots, r$, that is, the fully antisymmetrized basis, can be expanded on the original basis in Eq. (21),

$$|NiJT\rangle = \sum_{\dots} \langle (nlsjt; \mathcal{N}\mathcal{L}\mathcal{S}\mathcal{T}\mathcal{J}) | |NiJt\rangle | (nlsjt; \mathcal{N}\mathcal{L}\mathcal{S}\mathcal{T}\mathcal{J})JT \rangle \quad (35)$$

where $N = 2n + l + 2\mathcal{N} + \mathcal{L}$ and $\langle (nlsjt; \mathcal{N}\mathcal{L}\mathcal{S}\mathcal{T}\mathcal{J}) | |NiJt\rangle$ is the fractional parentage coefficient. This practically means that all calculations can be done in the non-antisymmetrized basis, and then converted to the antisymmetrized one such that the expectation value of an operator O is

$$\langle NiJT | O | N'i'JT \rangle = \sum \langle (nlsjt; \mathcal{N}\mathcal{L}\mathcal{S}\mathcal{T}\mathcal{J}) | |NiJt\rangle \langle (n'l's'j't'; \mathcal{N}'\mathcal{L}'\mathcal{S}'\mathcal{T}'\mathcal{J}') | |N'i'Jt\rangle \times \\ \langle (nlsjt; \mathcal{N}\mathcal{L}\mathcal{S}\mathcal{T}\mathcal{J})JT | O | (n'l's'j't'; \mathcal{N}'\mathcal{L}'\mathcal{S}'\mathcal{T}'\mathcal{J}')JT \rangle \quad (36)$$

Note that the procedure we described above relies on the antisymmetrization of the two-particle subsystem. If we generalize this, as we will do below, finding the fully antisymmetric basis for an A -particle system will require having a basis antisymmetric on the indices of the $A - 1$ -particle system. For the three nucleon system this was trivial, but as we increase the particle number this becomes inefficient because, e.g., to get a fully antisymmetric basis for the four-particle system we need the fully antisymmetric basis of the 3-particle system. In other words, to solve the for-particle problem we need to partially solve the three-particle problem first, and so on. We will see this now explicitly for the A -particle system.

First recall that the Jacobi coordinate ξ_n is proportional to the distance between particle $n + 1$ and the COM of the particles $1, 2, \dots, n$. Crucially, the coordinate of particle $n + 1$ does not enter any of the coordinates ξ_i with

$i = 1, 2, \dots, n - 1$. Hence, for the three-particle system, only ξ_2 described the motion of the 3 particle. Similarly, in a A -particle system, ξ_i with $i = 1, 2, \dots, A - 2$ describe the motion of particles up to particle $A - 1$. The motion of particle A is measured from the COM of particles $i = 1, 2, \dots, A - 1$ and it's described by coordinate ξ_{A-1} . Hence, in complete analogy to the 3 particle system, the state of the A particle system will be described by that of an antisymmetrized state of a cluster of $A - 1$ particles and a state of the A -th particle; the former will depend on coordinates $\xi_1, \xi_2, \dots, \xi_{A-2}$ and the latter will depend on coordinate ξ_{A-1} :

$$|\alpha\rangle_A = |(N_{A-1}iJ_{A-1}T_{A-1}; \mathcal{N}_{A-1}\mathcal{L}_{A-1}\mathcal{S}_{A-1}\mathcal{J}\tau_{A-1})JT\rangle, \quad (37)$$

where as before, $\mathcal{N}_{A-1}\mathcal{L}_{A-1}\mathcal{S}_{A-1}\mathcal{J}\tau_{A-1}$ are the quantum numbers of the A -th nucleon and $N_{A-1}iJ_{A-1}T_{A-1}$ are the quantum numbers of the fully antisymmetrized state of the other $A - 1$ nucleons. For the 3 nucleon system, this corresponds to the two-particle state $|kljt\rangle$ while for the 4 nucleon system, this is the fully antisymmetrized state in Eq. (??): again, and now explicitly, we need to know the eigenvectors of the antisymmetrizer $\mathcal{A}_3$ to construct the basis of the 4 nucleon system. Even having those, the state in Eq. (37) is antisymmetric in the particle labels $1, 2, \dots, A - 1$, but not in the last particle's label, A . As we did for the three-particle system we now have to find the fully antisymmetric states by diagonalizing the four-particle antisymmetrizer, $\mathcal{A}_4$, in the partially antisymmetric basis of Eq. (37), on which it is

$${}_A\langle\alpha'|\mathcal{A}_A|\alpha\rangle_A = \frac{1}{A} [1 - (A - 1) {}_A\langle\alpha'|P_{AA-1}|\alpha\rangle_A] \quad (38)$$

and the matrix element ${}_A\langle\alpha'|P_{AA-1}|\alpha\rangle_A$ can then be calculated with a generalization of Eq. (34). It should be clear now how this can get cumbersome to implement and more importantly expensive to calculate. In practice, only systems up to $A = 4$ are treated on this basis. For heavier systems the so-called Slater Determinant basis is used instead, which we describe below.

3.0.2 The many-body basis with Cartesian coordinates (*Slater Determinant Basis*)

In the previous section we saw how the shortcomings of the so-called Jacobi basis: while it is more natural for the description of self-bound systems of A particles, they become prohibitively inefficient for heavier systems. For this reason, in practice, a Cartesian coordinate basis is used through which one can leverage the efficiency of the second quantization to generate the basis.

The Slater Determinant basis is constructed out of single-particle HO states

$$|\alpha\rangle = |n_\alpha l_\alpha j_\alpha t_\alpha; b\rangle \quad (39)$$

where $\beta = \sqrt{\hbar/(m\omega)}$ is the oscillator length, that is, the length scale associated with the HO used to generate the basis; as such, it characterizes *the entire basis* (it doesn't have a i index). The corresponding wavefunctions are,

$$\begin{aligned} \phi_{nlsj_z t_z}(\mathbf{r}; \beta) &= \langle \mathbf{r} | nlsj_z t_z; b \rangle = R_{nl}(\mathbf{r}; b) \sum_{l_z s_z} \left[C_{l_z s_z j_z}^{lsj} Y_{l_z}(\hat{\mathbf{r}}) \chi_{s_z}(s) \right] \eta_{t_z}(t), \\ \text{with energy } E_{nl} &= \hbar\omega \left(2n + l + \frac{3}{2} \right) \end{aligned} \quad (40)$$

While Eq. (40) contains an explicit dependence on isospin projection, in practice one generates separate spin-orbitals ϕ_{nlj_z} for neutrons and for protons, as we'll explain in more detail below.

For a A -particle system, a set of A single particle states form the basis in Eq. (40) can then be combined to make many-particle states in the form of a Slater Determinant, labeled by the set of single-particle quantum numbers,

$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_A)$ similarly to what was discussed in Lecture 1. These many nucleon states will then be

$$\Phi_\alpha(\mathbf{R}) = \langle \mathbf{R} | \alpha \rangle = \frac{1}{\sqrt{A! \prod_{i=1}^A n_{\alpha_i}}} \begin{vmatrix} \phi_{\alpha_1}(\mathbf{r}_1) & \phi_{\alpha_2}(\mathbf{r}_1) & \cdots & \phi_{\alpha_A}(\mathbf{r}_1) \\ \phi_{\alpha_1}(\mathbf{r}_2) & \phi_{\alpha_2}(\mathbf{r}_2) & \cdots & \phi_{\alpha_A}(\mathbf{r}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{\alpha_1}(\mathbf{r}_A) & \phi_{\alpha_2}(\mathbf{r}_A) & \cdots & \phi_{\alpha_A}(\mathbf{r}_A) \end{vmatrix}, \quad (41)$$

where n_i is the degeneracy of the single-particle orbital i . As already described in Lecture 1, the set of states defined in Eq. (41) form a many-body basis, as long as, the states defined in Eq. (40) form a single-particle basis (which they do). In more technical terms, Eq. (41) presents a prescription on how to generate a complete basis of the A -particle Fock space $\mathcal{F}_A$ from a complete basis of the 1-particle Fock space $\mathcal{F}_1$. Note that in this approach one constructs fully antisymmetric states immediately without having to diagonalize the antisymmetrizer (and two away 2/3 of the resulting eigenstates). In practice, even though the A -body basis is constructed from the $A - 1$ -body basis, the construction is trivial and equivalent to adding a column and a row to the matrix in Eq. (41). Hence, creating a fully antisymmetric basis for a A -particle system is not a bottleneck anymore.

Now given the complete many-body basis $\{\alpha\}$, the exact ground (and any excited) state of the Hamiltonian can be written as

$$|\Psi\rangle = \sum_{\alpha} c_{\alpha} |\alpha\rangle, \quad \text{with energy } E = \sum_{\alpha\alpha'} c_{\alpha'}^* c_{\alpha} \langle \alpha' | H | \alpha \rangle \quad (42)$$

The extremely simple form of this equation gives these methods the name *expansion methods*. By use of the Lawson projection term introduced in Eq. (??), one can push parametrically to infinity the energy required to excite the center of mass, which means that for large enough values of β , the COM wavefunction is always found with a wavefunction described by no harmonic oscillator quanta, or,

$$\langle \mathbf{R} | \alpha \rangle = \langle \Xi | \alpha \rangle \Phi_0(\xi_0) \quad (43)$$

where Φ_0 represents the many-body HO state with excitations equal to $0\hbar\omega$. In practice the diagonalization of the Hamiltonian, be it in the Jacobi basis or the SD basis, is done in a truncated space. This truncation must be done with care because the factorization of the CM in the calculated ground state depends on it. The right way to do it is to keep all many-body states for which $\sum_i^N (2n_i + l_i) \leq N_0(N) + N_{\max}$, where $N_0(N)$ is often referred to as the lowest Pauli configuration and originates from naively filling up the HO single-particle states with N particles, e.g., $N_0(3) = N_0(4) = 0$, $N_0(5) = 2$, etc. Then $N_{\max}\hbar\omega$ becomes the maximally allowed HO excitation energy. As such both $N_{\max}$ and $\hbar\omega$ are parameters controlling this truncation but they are also variational parameters, in other words, the energy should be converged and minimized. As such $N_{\max}$ is increased to convergence of the spectrum and $\hbar\omega$ is adjusted for minimization.

4 Similarity Renormalization Group evolution

In Lecture 3 we discussed about the various descriptions of the nuclear force. While many successful descriptions are currently available, some are more practical than others. For example, potentials with strong short range components can induce short range correlations. An expansion method based on the HO orbitals needs configurations with very high excitations to describe short range correlations, which translates to expensive calculations with high $N_{\max}$ values. This problem can be softened with a set of methods that transform away these problematic parts of the nuclear Hamiltonian. In this section we will review a modern standard approach which is the Similarity Renormalization Group (SRG) method.

The (SRG) method uses a continuous set of unitary transformations to provide an equivalent Hamiltonian whose strong short-range components have been redistributed to longer scales. Since the transformation is unitary, all

physical properties of the Hamiltonian should in principle be conserved. This type of evolution can be seen as a continuous reduction of the resolution scale of the interactions: the more we transform the more we lose information from the short-range dynamics, reconstructing the longer-range parts to incorporate them. This inevitably gives rise to many-body forces similarly to how reducing our resolution to ignore the size of the nucleon gives rise to many-body forces, as we discussed in Lecture 3; in this case we will see this play out explicitly.

The method of the SRG transformation amounts to defining a Hamiltonian

$$H_s = U(s) H U(s)^\dagger , \quad (44)$$

for $s \in (0, \infty)$ where the unitary operator $U(s)$ must be unitary $U(s)U^\dagger(s) = 1$ and appropriately chosen. Unitary transformation can be defined in terms of a anti-hermitian generator,

$$U(s) = \mathcal{T} \exp \left[\int_s^0 \eta(s') ds' \right] , \quad (45)$$

where the ordered exponential $\mathcal{T} \exp$ specifies that operators should be ordered in increasing energy scale s and makes sure that the exponential of an operator's integral is uniquely defined. Since U is unitary, the anti-Hermitian generator operator can be written as

$$\eta(s) = \frac{dU(s)}{ds} U^\dagger(s) . \quad (46)$$

This allows the continuous transformation in Eq. (44) to be written as a differential equation,

$$\frac{dH_s}{ds} = [H_s, \eta(s)] , \quad (47)$$

which is to be integrated with the boundary conditions $H(0) = H$, that is, before transforming anything the Hamiltonian was the original, or *bare* Hamiltonian. Often a simple choice is made for a generator,

$$\eta(s) = [T_{\text{rel}}, H] , \quad (48)$$

where T_{rel} is the first term in Eq. (17). Now it is straightforward to see that even with we started with only two-body forces, the evolution would give rise to higher-body forces. Take for example

$$H = T_{\text{rel}} + V_2 = \sum_{ij=1}^N t_{ij} c_i^\dagger c_j + \sum_{ijkl=1}^N v_{ij,kl}^{(2)} c_i^\dagger c_j^\dagger c_l c_k \quad (49)$$

and evaluate $[T_{\text{rel}}, V_2]$ which will generate N -body forces. In practice Eq. (47) is solved as a differential equation for each matrix element of the Hamiltonian on the Jacobi basis where the matrix elements are evaluated already. Then the dependence of the calculation on the final value of s is taken as an indicator of the unitarity of the calculation, or lack thereof caused by neglected induced higher-body interactions.

5 Lancos algorithm for diagonalization

The preceding sections cast the solution of the many-particle Schrödinger's equation into a linear algebra problem: the spectrum of the Hamiltonian in Eq. (17) can be determined by diagonalizing the matrix

$$H_{\alpha'\alpha} = \langle \alpha' | H | \alpha \rangle . \quad (50)$$

While this looks like a trivial problem it can be a prohibitively hard one. In practice, the number of states needed in the Cartesian basis can go up to 10^9 which then becomes the dimension of $H_{\alpha\alpha'}$. If done naively, the diagonalization of such a matrix is doomed to fail. Many approaches that tackle this problem exist and here we'll describe a powerful and, hence, common one: Lanczos's algorithm for matrix diagonalization.

Lanczos' algorithm is an iterative method that actually aims at turning a Hermitian matrix $H_{\alpha'\alpha}$ into a tri-diagonal form from which the matrix's spectrum of eigenvalues can be read. Its very attractive feature is that it can be seen as systematically improving to convergence on the eigenvalue spectrum with lower eigenvalues converging first. This means that for a matrix with dimension 10^6 , one can avoid spending the resources to calculate the 10^4 -th eigenvalue and its corresponding eigenvector to the same precision as the first and second eigenvalues. Hence, Lanczos's algorithm is ideal for calculation's of the first few eigenstates of a matrix, or in physical terms, the ground and first few excited states of a Hamiltonian. More importantly, being an iterative method, it provides a systematically improvable approximation to the Hamiltonian's eigenstates. The algorithm starts from a "random" seed, a normalized state, namely, $|\eta_0\rangle$, also known as the *pivot* which should overlap with the desired part of the spectrum, i.e., the ground state the first few excited states. This and later statements might echo similar ones made during the QMC lecture; this is not coincidence. Then a second state is generated as

$$|\tilde{\eta}_1\rangle = H|\eta_0\rangle - a_0|\eta_0\rangle, \quad \text{where } a_0 = \langle\eta_0|H|\eta_0\rangle. \quad (51)$$

Note that $|\tilde{\eta}_1\rangle$ is *not* normalized but rather its magnitude $\sqrt{\langle\tilde{\eta}_1|\tilde{\eta}_1\rangle}$ tells us how far $|\eta_0\rangle$ is from being an eigenvector: if $|\eta_0\rangle$ were an eigenvector then $H|\eta_0\rangle = E_0|\eta_0\rangle = a_0|\eta_0\rangle$ and $\sqrt{\langle\tilde{\eta}_1|\tilde{\eta}_1\rangle} = 0$. So we will name $\beta_0 = \sqrt{\langle\tilde{\eta}_1|\tilde{\eta}_1\rangle}$ and rewrite Eq. (51) for the normalized vector $|\eta_1\rangle = |\tilde{\eta}_1\rangle/\beta_0$:

$$\beta_0|\eta_1\rangle = H|\eta_0\rangle - a_0|\eta_0\rangle, \quad \text{where } a_0 = \langle\eta_0|H|\eta_0\rangle. \quad (52)$$

Note the probability of starting with a random pivot that is an eigenstate of the Hamiltonian in a problem of large dimensionality, such as the ones we want to solve, is tiny. By projecting Eq. (52) onto $|\eta_0\rangle$ we see that $|\eta_1\rangle$ is orthogonal to $|\eta_0\rangle$:

$$\langle\eta_0|\tilde{\eta}_1\rangle = \langle\eta_0|H|\eta_0\rangle - a_0\langle\eta_0|\eta_0\rangle = 0 \quad (53)$$

We then take vector $|\tilde{\eta}_1\rangle$, or rather its normalized version, $|\eta_1\rangle = |\tilde{\eta}_1\rangle/\beta_1$ which is now orthonormal to $|\eta_0\rangle$, and treat it the same way: we use it to generate a third vector

$$|\tilde{\eta}_2'\rangle = H|\eta_1\rangle - a_1|\eta_1\rangle, \quad \text{where } a_1 = \langle\eta_1|H|\eta_1\rangle. \quad (54)$$

Similarly $|\tilde{\eta}_2'\rangle$ is not normalized with its magnitude telling of the distance of $|\eta_1\rangle$ from being an eigenstate and, as with $|\tilde{\eta}_1\rangle$ and $|\eta_0\rangle$, $|\tilde{\eta}_2'\rangle$ is orthogonal to $|\eta_1\rangle$. This won't change if we were to add to the RHS of Eq. (54) states that are orthogonal to $|\eta_1\rangle$. Since the intention is to keep iterating in a similar manner we would also like for $|\tilde{\eta}_2'\rangle$ to be orthogonal to the original $|\eta_0\rangle$. Projecting Eq. (54) onto $|\eta_0\rangle$ we get:

$$\langle\eta_0|\tilde{\eta}_2'\rangle = \langle\eta_0|H|\eta_1\rangle - a_1\langle\eta_0|\eta_1\rangle = \langle\eta_0|H|\eta_1\rangle \quad (55)$$

that is, $|\tilde{\eta}_2'\rangle$ and $|\eta_0\rangle$ are not orthogonal. Projecting Eq. (52) on state $|\eta_1\rangle$ we get

$$b_0 = \langle\eta_1|H|\eta_0\rangle = \langle\eta_0|H|\eta_1\rangle \quad (56)$$

where we also used that H is a Hermitian matrix. Hence we can add $b_0|\eta_0\rangle$ to the RHS of Eq. (54) and make $|\tilde{\eta}_2'\rangle$ orthogonal to $|\eta_0\rangle$, while keeping it orthogonal to $|\eta_1\rangle$ (because $\langle\eta_0|\eta_1\rangle = 0$). As you might have picked up on, this procedure is very similar to the Gram-Schmidt orthogonalization. Hence, we define the state

$$|\tilde{\eta}_2\rangle = H|\eta_1\rangle - a_1|\eta_1\rangle - b_0|\eta_0\rangle. \quad (57)$$

and as before we normalize writing $|\eta_2\rangle = |\tilde{\eta}_2\rangle / \sqrt{\langle\tilde{\eta}_2|\tilde{\eta}_2\rangle}$, and similarly as before we'll name $b_1 = \sqrt{\langle\tilde{\eta}_2|\tilde{\eta}_2\rangle}$ as it measures how far $|\eta_1\rangle$ is from being an eigenstate of H . We finally arrive at:

$$b_1 |\eta_2\rangle = H |\eta_1\rangle - a_1 |\eta_1\rangle - b_0 |\eta_0\rangle . \quad (58)$$

We now have a pattern and we can define: given a state $|\eta_i\rangle$ we get the state $|\eta_{i+1}\rangle$ as

$$b_i |\eta_{i+1}\rangle = H |\eta_i\rangle - a_i |\eta_i\rangle - b_{i-1} |\eta_{i-1}\rangle . \quad (59)$$

where a_i is the Hamiltonian's expectation value at state $|\eta_i\rangle$ and b_i is a normalization, or,

$$a_i = \langle\eta_i| H |\eta_i\rangle , \quad b_i = \|H |\eta_i\rangle - a_i |\eta_i\rangle - b_{i-1} |\eta_{i-1}\rangle\|^2 \quad (60)$$

Another way to see this construction is by starting with state $|\eta_i\rangle$ and getting $|\eta_{i+1}\rangle$ by acting on it with the Hamiltonian to get state $H |\eta_i\rangle$, then subtracting $a_i |\eta_i\rangle$ from it to make it orthogonal to $|\eta_i\rangle$ and $b_{i-1} |\eta_{i-1}\rangle$ to make it orthogonal to $|\eta_{i-1}\rangle$. Repeating this M times creates a set of vectors $\{|\eta_i\rangle\}$ that are orthogonal to each other and so the Hamiltonian on that basis assumes a tri-diagonal form given entirely by the coefficients a_i and b_i . This is a compression of the original Hamiltonian into a lower dimensional space of dimension M and, of course, when $M = N_{\text{SD}}$ the compression vanishes. Correspondingly the eigenvalues and eigenvectors of the compressed Hamiltonian match those of the full one when $M = N_{\text{SD}}$ but otherwise they constitute *systematically improvable* approximations to them. Crucially, the lower parts of the spectrum converge faster than the higher ones and one doesn't need to spend the extra labour associated with calculating highly excited states when interested in the ground state and/or its low excitations.